Classical Field Theory — Exercise Sheet 21

Vector mass matrices and longitudinal energy

Solutions

Use the signature $(+,-,-,-)$ and canonically normalised real scalar kinetic coefficient $1/2$. The questions concern quadratic theories near a fixed vacuum. They make no assertion about a global unitary gauge.

Exercise 21.1 (An orbit determines a vector mass matrix). Let a compact Lie group act orthogonally on $\mathbb{R}^N$, and choose a basis of its Lie algebra, represented by skew real generators T_a, $1 \le a \le r$. Let u_0 be a vacuum of an invariant C^2 potential. Fix $g > 0$, and give the scalar the kinetic term

$$\frac{1}{2}(D_\mu\phi)\cdot(D^\mu\phi), \qquad D_\mu\phi = \partial_\mu\phi + gA^a_\mu T_a\phi.$$

The vector kinetic terms are canonical at the quadratic level.

(a) Derive the vector squared-mass matrix at $\phi = u_0$. Prove that it is nonnegative and identify its kernel and rank. Include generators which act trivially in the representation.

(b) For the three-dimensional rotation representation, take

$$T_a w = e_a \times w, \qquad u_0 = (v/\sqrt{2}, v/\sqrt{2}, 0), \qquad v > 0.$$

Compute the full matrix, its orthonormal eigenvectors and the vector masses. Which gauge-field combination remains massless?

Solution. For $\phi = u_0 + h$, the part of the scalar kinetic term quadratic in the vector alone is

$$\frac{1}{2}g^2A^a_\mu A^{b\mu}(T_a u_0)\cdot(T_b u_0) = \frac{1}{2}A^a_\mu \mathsf{M}_{ab}A^{b\mu}.$$

The remaining quadratic terms include the scalar kinetic and scalar–vector mixing; they do not alter this coefficient matrix. For a real array z its quadratic form is

$$z^a\mathsf{M}_{ab}z^b = g^2\left|\sum_a z^aT_a u_0\right|^2 \ge 0.$$

It vanishes exactly when the displayed orbit direction vanishes. Thus the kernel is the Lie algebra of the stabiliser, including any ineffective generators. Lecture 17 identifies the rank of the orbit map as $\dim G - \dim H$, giving the asserted rank.

For the rotation representation the cross-product identity gives

$$\mathsf{M} = g^2(v^2 I - u_0 u_0^{\mathsf{T}}).$$

Substitution gives

$$\mathsf{M} = g^2v^2\begin{pmatrix} 1/2 & -1/2 & 0 \\ -1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

An orthonormal eigenbasis in generator space is

$$z_0 = \frac{1}{\sqrt{2}}(1,1,0), \qquad z_1 = \frac{1}{\sqrt{2}}(1,-1,0), \qquad z_2 = (0,0,1).$$

Their squared masses are respectively 0, g^2v^2 and g^2v^2. Canonical vector kinetic terms make their positive mass values gv. The unbroken generator rotates about u_0; its gauge component is $z_0 \cdot A$. The two orthogonal generator directions move the vacuum.

Exercise 21.2 (The charged and neutral electroweak combinations). The following bosonic representation data are given. A complex doublet $\Phi \in \mathbb{C}^2$ has kinetic term $\frac{1}{2}(D_\mu\Phi)^\dagger D^\mu\Phi$, potential $\lambda(\Phi^\dagger\Phi - v^2)^2/8$, and vacuum $\Phi_0 = (0,v)^\mathsf{T}$. Here $g, g', v, \lambda > 0$ and

$$D_\mu = \partial_\mu - \mathrm{i}\left(gW_\mu^a\frac{\sigma_a}{2} + g'B_\mu\frac{I_2}{2}\right).$$

All four real vector fields have canonical quadratic Maxwell kinetic terms. The Pauli matrices and the hypercharge generator $Y = I_2/2$ specify the representation; no fermion sector is included.

(a) Find the full real squared-mass matrix in the order (W^1, W^2, W^3, B). Diagonalise its neutral block and define Z and the massless photon A_γ with unit kinetic normalisation. Define $W^\pm = (W^1 \mp \mathrm{i}W^2)/\sqrt{2}$ and find their common mass.

(b) Find the generator annihilating Φ_0 and the coefficient of its photon coupling in D_μ. Find the radial scalar mass and compare the quadratic number of physical modes in the uncoupled limit $g, g' \to 0$ with the broken quadratic description at positive couplings. Explain why W^+ and W^- do not count as two independent complex vectors.

This is the bosonic mass calculation of the electroweak model. For the original construction see Weinberg, *Physical Review Letters* **19** (1967), 1264–1266, equations (4), (7)–(12). The signs and scalar normalisation to use are those explicitly given above.

Solution. At the vacuum the vector part of the derivative is

$$D_\mu\Phi_0 = -\frac{\mathrm{i}v}{2}\begin{pmatrix} g(W_\mu^1 - \mathrm{i}W_\mu^2) \\ -gW_\mu^3 + g'B_\mu \end{pmatrix}.$$

Its squared norm, multiplied by the kinetic coefficient $1/2$, gives

$$L_{\text{mass}} = \frac{v^2}{8}\left[g^2(W_\mu^1 W^{1\mu} + W_\mu^2 W^{2\mu}) + (-gW_\mu^3 + g'B_\mu)(-gW^{3\mu} + g'B^\mu)\right].$$

Thus the real squared-mass matrix is

$$\mathsf{M} = \frac{v^2}{4}\begin{pmatrix} g^2 & 0 & 0 & 0 \\ 0 & g^2 & 0 & 0 \\ 0 & 0 & g^2 & -gg' \\ 0 & 0 & -gg' & g'^2 \end{pmatrix}.$$

The orthonormal neutral eigenvectors are $(g,-g')/g_Z$ and $(g',g)/g_Z$, with eigenvalues $g_Z^2v^2/4$ and zero. Their orthogonal change of variables preserves the quadratic kinetic terms and defines

$$Z = \frac{gW^3 - g'B}{g_Z}, \qquad A_\gamma = \frac{g'W^3 + gB}{g_Z}.$$

Their masses are $g_Z v/2$ and zero. The two charged real directions have eigenvalue $g^2v^2/4$. Combining them into $W^\pm = (W^1 \mp \mathrm{i}W^2)/\sqrt{2}$ gives $m_W = gv/2$.

The generator $Q_{\mathrm{em}} = \sigma_3/2 + Y$ annihilates $(0, v)^{\mathsf{T}}$. The inverse neutral rotation is

$$W^3 = (gZ + g'A_\gamma)/g_Z, \qquad B = (-g'Z + gA_\gamma)/g_Z.$$

Consequently

$$gW^3\frac{\sigma_3}{2} + g'BY = \frac{g^2\sigma_3/2 - g'^2Y}{g_Z}Z + \frac{gg'}{g_Z}Q_{\mathrm{em}}A_\gamma.$$

Thus its photon coupling is $e = gg'/g_Z$.

Along the radial line $\Phi = (0, v+h)^{\mathsf{T}}$, the kinetic term is $\frac{1}{2}(\partial h)^2$ and the quadratic potential is $\frac{1}{2}\lambda v^2 h^2$. Hence $m_h = \sqrt{\lambda}\, v$. The vacuum orbit of the supplied action has three independent tangents: the mass matrix has rank three, and those generator directions move Φ_0. The remaining scalar direction is radial. In the uncoupled limit $g, g' \to 0$, there are four massless real vectors with two polarisations each and four real scalar components, giving $8 + 4 = 12$. The scalar potential has one radial curvature λv^2 and three orbit zeros, so the scalar count includes one massive and three massless components. After the split there are three massive real vectors with three polarisations each, the photon with two, and the radial scalar with one, giving $9 + 2 + 1 = 12$. Since W^1 and W^2 are real, $W^- = \overline{W^+}$. They encode two real massive vectors, not two freely specifiable complex vectors. This count concerns the local quadratic theory only.

Exercise 21.3 (The positive reduced Proca Hamiltonian). Use Lecture 21's variables $B = -\beta\,\mathrm{d}t + \mathbf{b}\cdot\mathrm{d}x$, mass $m > 0$ and spatial momentum $\mathbf{\Pi}$. Take real Schwartz data on $\mathbb{R}^3$. Before elimination the Hamiltonian is

$$H_B = \int_{\mathbb{R}^3}\left[\frac{1}{2}|\mathbf{\Pi}|^2 + \frac{1}{2}|\operatorname{curl}\mathbf{b}|^2 + \frac{1}{2}m^2|\mathbf{b}|^2 + \beta\operatorname{div}\mathbf{\Pi} - \frac{1}{2}m^2\beta^2\right]\mathrm{d}^3x.$$

Its scalar constraints are $\pi_\beta = 0$ and $\operatorname{div}\mathbf{\Pi} - m^2\beta = 0$.

(a) Eliminate the scalar pair and derive the reduced Hamiltonian and both spatial Hamilton equations with the remaining canonical form. Prove positivity and show that a zero-energy real datum is zero.

(b) At a nonzero Fourier vector k, split the spatial amplitudes into longitudinal and transverse parts. Find the frequencies and a canonical normalisation of each real longitudinal Fourier quadrature. Treat $k = 0$ separately. Why does the energy formula not have a regular mass-zero limit for fixed arbitrary longitudinal momentum?

Individual Fourier modes here analyse the multiplier; they are not claimed to be finite-energy plane-wave fields on $\mathbb{R}^3$.

Solution. The terms depending on β can be written

$$\beta\operatorname{div}\mathbf{\Pi} - \frac{1}{2}m^2\beta^2 = -\frac{1}{2}m^2\left(\beta - \frac{\operatorname{div}\mathbf{\Pi}}{m^2}\right)^2 + \frac{|\operatorname{div}\mathbf{\Pi}|^2}{2m^2}.$$

The secondary constraint sets the square to zero. Since $\pi_\beta = 0$, the pullback of the scalar canonical term vanishes, leaving the spatial canonical form. The reduced Hamiltonian is therefore

$$H_{\mathrm{red}} = \frac{1}{2}\int\left[|\mathbf{\Pi}|^2 + |\operatorname{curl}\mathbf{b}|^2 + m^2|\mathbf{b}|^2 + m^{-2}|\operatorname{div}\mathbf{\Pi}|^2\right]\mathrm{d}^3x.$$

Its remaining bracket is the spatial canonical bracket.

Schwartz integration by parts in its variation gives

$$(H_{\mathrm{red}})_{\boldsymbol{\Pi}} = \boldsymbol{\Pi} - m^{-2}\nabla \operatorname{div} \boldsymbol{\Pi}, \qquad (H_{\mathrm{red}})_{\mathbf{b}} = \operatorname{curl}\operatorname{curl} \mathbf{b} + m^2 \mathbf{b}.$$

Hamilton's equations therefore give

$$\mathbf{b}_t = \boldsymbol{\Pi} - m^{-2}\nabla \operatorname{div} \boldsymbol{\Pi}, \qquad \boldsymbol{\Pi}_t = -\operatorname{curl}\operatorname{curl} \mathbf{b} - m^2 \mathbf{b}.$$

Every energy term is nonnegative. If the energy is zero, the momentum norm and m^2 times the spatial-field norm both vanish, so the datum is zero.

For $k \neq 0$, the transverse symbols obey

$$\partial_t \widehat{\mathbf{b}}_T = \widehat{\boldsymbol{\Pi}}_T, \qquad \partial_t \widehat{\boldsymbol{\Pi}}_T = -(|k|^2 + m^2)\widehat{\mathbf{b}}_T.$$

The longitudinal symbols instead obey

$$\partial_t b_L = \left(1 + \frac{|k|^2}{m^2}\right) \Pi_L, \qquad \partial_t \Pi_L = -m^2 b_L.$$

A second derivative gives frequency squared $|k|^2 + m^2$ in both cases. For one real quadrature the longitudinal Hamiltonian is $\frac{1}{2}[(\nu^2/m^2)\Pi_L^2 + m^2 b_L^2]$. Put $Q_L = (m/\nu)b_L$ and $P_L = (\nu/m)\Pi_L$. This reciprocal rescaling preserves $\mathrm{d}b_L \wedge \mathrm{d}\Pi_L$ and turns it into $\frac{1}{2}(P_L^2 + \nu^2 Q_L^2)$. At $k = 0$ no direction is selected as longitudinal; all three spatial axes obey frequency m.

For a fixed nonzero longitudinal momentum at $k \neq 0$, the term $|k|^2|\Pi_L|^2/(2m^2)$ diverges as $m \downarrow 0$. The scalar constraint elimination itself divides by m^2. It therefore cannot describe Maxwell's mass-zero gauge reduction by simply substituting $m = 0$. This agrees with the singular phase-variable replacement in Lecture 21.